PROBLEM 1.1

Classification of Ten Fundamental Differential Equations

Original statement · ProblemsCh1.pdf

Taxonomy • Order • Linearity

PHASE 1: PROBLEM STATEMENT & HYPOTHESES

Problem statement: Classify the following equations as ordinary or partial; give their order; and state whether they are linear or nonlinear. In each case identify the dependent and independent variables:

1. (i) **Bessel's equation** (ν is a parameter): $x^2y'' + xy' + (x^2 - \nu^2)y = 0$
2. (ii) **Burgers' equation** (ν is a parameter): $\frac{\partial u}{\partial t} - \nu\frac{\partial^2 u}{\partial x^2} + u\frac{\partial u}{\partial x} = 0$
3. (iii) **Duffing's equation** (m, k, a, b are parameters): $m\ddot{x} + kx = ax - bx^3$
4. (iv) $\frac{dy}{dt} = t - y^2$
5. (v) **The wave equation** (c is a parameter): $\frac{\partial^2 y}{\partial t^2} = c^2\frac{\partial^2 y}{\partial x^2}$
6. (vi) **Newton's law of cooling** (k is a parameter and $T_A(t)$ is a specified function): $\frac{dT}{dt} = -k(T - T_A(t))$
7. (vii) **The logistic population model** (k is a parameter): $\frac{dp}{dt} = kp(1 - p)$
8. (viii) **Newton's second law** for a particle of mass m moving in a potential $V(x)$: $m\ddot{x} = -V'(x)$
9. (ix) **The coupled equations:** $\dot{x} = x(4 - 2x - y), \quad \dot{y} = y(9 - 3x - 3y)$
10. (x) $\frac{d\mathbf{x}}{dt} = A\mathbf{x}$, where $\mathbf{x}$ is an n-component vector and A is an $n \times n$ matrix.

PHASE 2: FUNDAMENTAL FORMULATION & CRITERIA

METHOD OF THE CLASS NOTES [NOTES, P. 2]

- **ODE / PDE:** the unknown function depends on only one independent variable (ODE) or on more than one (PDE).
- **Order:** the order of the highest derivative.
- **System:** two or more dependent variables.
- **Linear:** the ODE has the form $a_0(x)y^{(n)} + \cdots + a_n(x)y = b(x)$, with coefficients and right-hand side depending only on the independent variable. The notes test it by superposition (on the homogeneous part): if u_1, u_2 are solutions, $\alpha u_1 + \beta u_2$ must be a solution for all $\alpha, \beta \in \mathbb{R}$.

PHASE 3: STEP-BY-STEP MATHEMATICAL DERIVATION

(i) Bessel's equation: $x^2y'' + xy' + (x^2 - \nu^2)y = 0$
Independent variable: x (single $\implies$ **ODE**). Dependent variable: $y(x)$.
Derivatives present: y' and y''. Highest derivative is order 2 $\implies$ **Order 2**.
Linearity: Matches $a_2(x)y'' + a_1(x)y' + a_0(x)y = b(x)$ with $a_2(x) = x^2, a_1(x) = x, a_0(x) = x^2 - \nu^2, b(x) = 0$. All y, y', y'' appear linearly $\implies$ **Linear**.

(ii) Burgers' equation: $\frac{\partial u}{\partial t} - \nu\frac{\partial^2 u}{\partial x^2} + u\frac{\partial u}{\partial x} = 0$
Independent variables: t, x (two $\implies$ **PDE**). Dependent variable: $u(x, t)$.
Highest derivative: $\frac{\partial^2 u}{\partial x^2}$ $\implies$ **Order 2**.
Linearity: The convective acceleration term $u\frac{\partial u}{\partial x}$ is a product of the dependent variable and its derivative $\implies$ **Nonlinear**.

(iii) Duffing's equation: $m\ddot{x} + kx = ax - bx^3$
Overdot signifies time derivative: $\ddot{x} = \frac{d^2x}{dt^2}$. Independent variable: t $\implies$ **ODE**. Dependent variable: $x(t)$.
Highest derivative: $\ddot{x}$ $\implies$ **Order 2**.
Linearity: Rearranging gives $m\ddot{x} + (k - a)x + bx^3 = 0$. If $b \neq 0$, the cubic power x^3 violates linearity $\implies$ **Nonlinear**.
Superposition test of the notes. If x_1, x_2 are solutions and $x = \alpha x_1 + \beta x_2$, linearity of differentiation gives $m\ddot{x} + (k - a)x = \alpha(-bx_1^3) + \beta(-bx_2^3)$, so $m\ddot{x} + (k - a)x + bx^3 = b\big[(\alpha x_1 + \beta x_2)^3 - \alpha x_1^3 - \beta x_2^3\big]$.
With $\alpha = 2, \beta = 0$ this equals $b(8 - 2)x_1^3 = 6bx_1^3 \neq 0$: superposition fails.

(iv) Riccati Type: $\frac{dy}{dt} = t - y^2$
Independent variable: t $\implies$ **ODE**. Dependent variable: $y(t)$. Highest derivative: $\frac{dy}{dt}$ $\implies$ **Order 1**.
Linearity: The term y^2 is quadratic in the dependent variable $\implies$ **Nonlinear**.

(v) Wave Equation: $\frac{\partial^2 y}{\partial t^2} = c^2\frac{\partial^2 y}{\partial x^2}$
Independent variables: t, x $\implies$ **PDE**. Dependent variable: $y(x, t)$ (displacement).
Highest derivative: $\frac{\partial^2 y}{\partial t^2}$, $\frac{\partial^2 y}{\partial x^2}$ $\implies$ **Order 2**.
Linearity: $\frac{\partial^2 y}{\partial t^2} - c^2\frac{\partial^2 y}{\partial x^2} = 0$ is a homogeneous linear differential operator $\implies$ **Linear**.

(vi) Newton's Law of Cooling: $\frac{dT}{dt} = -k(T - T_A(t))$
Independent variable: $t \implies$ **ODE**. Dependent variable: $T(t)$. Highest derivative: $\frac{dT}{dt} \implies$ **Order 1**.
Linearity: Rearranges to $\frac{dT}{dt} + kT = kT_A(t)$, matching $a_1(t)\dot{T} + a_0(t)T = b(t) \implies$ **Linear**.

(vii) Logistic Model: $\frac{dp}{dt} = kp(1 - p)$
Independent variable: $t \implies$ **ODE**. Dependent variable: $p(t)$. Highest derivative: $\frac{dp}{dt} \implies$ **Order 1**.
Linearity: Expanding yields $\frac{dp}{dt} - kp + kp^2 = 0$. The term kp^2 is nonlinear in $p \implies$ **Nonlinear**.

(viii) Particle in Potential: $m\ddot{x} = -V'(x)$
Independent variable: $t \implies$ **ODE**. Dependent variable: $x(t)$. Highest derivative: $\ddot{x} \implies$ **Order 2**.
Linearity: The force $-V'(x)$ is in general a nonlinear function of x (linear only if $V(x) = \frac{1}{2}\kappa x^2$) $\implies$ **Nonlinear**.

(ix) Coupled Competing Species: $\dot{x} = x(4 - 2x - y), \quad \dot{y} = y(9 - 3x - 3y)$
Independent variable: $t \implies$ **System of ODEs**. Dependent variables: $x(t), y(t)$. Highest derivative: 1st order $\implies$ **Order 1**.
Linearity: Expanding gives $\dot{x} = 4x - 2x^2 - xy$ and $\dot{y} = 9y - 3xy - 3y^2$, containing quadratic terms (x^2, y^2) and cross product (xy) $\implies$ **Nonlinear**.

(x) Vector Matrix System: $\frac{d\mathbf{x}}{dt} = A\mathbf{x}$
Independent variable: $t \implies$ **System of ODEs**. Dependent variables: $\mathbf{x}(t) \in \mathbb{R}^n$. Highest derivative: 1st order $\implies$ **Order 1**.
Linearity: Matrix multiplication is a linear operator: $A(c_1\mathbf{x}_1 + c_2\mathbf{x}_2) = c_1A\mathbf{x}_1 + c_2A\mathbf{x}_2 \implies$ **Linear**.

PHASE 4: FINAL RESULT & ANALYSIS

MASTER CLASSIFICATION MATRIX

Item	Equation	Type	Order	Linearity	Dependent	Independent
(i)	$x^2y'' + xy' + (x^2 - \nu^2)y = 0$	ODE	2	Linear	y	x
(ii)	$u_t - \nu u_{xx} + uu_x = 0$	PDE	2	Nonlinear	u	t, x
(iii)	$m\ddot{x} + kx = ax - bx^3$	ODE	2	Nonlinear	x	t
(iv)	$y' = t - y^2$	ODE	1	Nonlinear	y	t
(v)	$y_{tt} = c^2y_{xx}$	PDE	2	Linear	y	t, x
(vi)	$T' = -k(T - T_A(t))$	ODE	1	Linear	T	t
(vii)	$p' = kp(1 - p)$	ODE	1	Nonlinear	p	t
(viii)	$m\ddot{x} = -V'(x)$	ODE	2	Nonlinear	x	t
(ix)	$\dot{x} = x(4 - 2x - y), \dot{y} = y(9 - 3x - 3y)$	ODE System	1	Nonlinear	x, y	t
(x)	$\dot{\mathbf{x}} = A\mathbf{x}$	ODE System	1	Linear	$\mathbf{x}$	t